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To design web server and json data creation and parsing it using Node Js and exploration of npm commands.

Description:

The web server is developed using Node.js to handle client requests and responses efficiently. JSON is used for data storage and exchange, where data is dynamically created and parsed on the server side. Node.js processes the JSON data, extracts required information, and sends structured responses to the client, enabling smooth and organized communication between the server and applications and exploration on npm commands.

Node JS Concepts Used:

- **Node.js runtime environment** – Executes JavaScript on the server side
- **HTTP module** – Creates and runs a web server
- **Web server creation** – Using `http.createServer()` to handle client requests and responses
- **HTTP response handling** – Setting headers and sending responses using `writeHead()` and `end()`
- **Port listening** – Running the server on a specified port using `server.listen()`
- **File System (fs) module** – Reads external files from the system
- **JSON data handling** – Reading and processing JSON files
- **JSON parsing** – Converting JSON content into usable JavaScript objects
- **Asynchronous programming** – Using callbacks or promises for file reading and data processing
- **Error handling** – Managing errors during file reading and JSON parsing

Exploration of NPM commands:

- `npm init` - Create project + `package.json` (asks questions)
- `npm init -y` - Create project + `package.json` (default values)
- `npm install express` - Install Express package
- `npm remove <package>` - Uninstall a package
- `npm run <script>` - Run a script from `package.json`
- `npm list` - Show installed packages
- `npm start` - Run "start" script
- `npm stop` - Run "stop" script
- `npm test` - Run "test" script

Code:**Server.js:**

```
const http = require('http');
```

```

const fs = require('fs');

const server = http.createServer((req, res) => {
  fs.readFile('./data.json', 'utf8', (err, jsonString) => {
    if (err) {
      res.writeHead(500, {'Content-Type': 'text/plain'});
      res.end("Error reading data.json file...");
      return;
    }
    try {
      const data = JSON.parse(jsonString);
      res.writeHead(200, {'Content-Type': 'application/json'});
      res.end(JSON.stringify(data, null, 2));
    } catch (err) {
      res.writeHead(500, {'Content-Type': 'text/plain'});
      res.end("Error parsing JSON data.");
    }
  });
});

server.listen(3000, () => {
  console.log('Server is running at http://localhost:3000');
});

```

Parse.js:

```

const fs = require("fs");
const xml2js = require("xml2js");
const parser = new xml2js.Parser();
fs.readFile("students.xml", (err, data) => {
  if (err) throw err;
  parser.parseString(data, (err, result) => {
    if (err) throw err;
    console.log(result.students.student);
  });
});

```

Data.json:

```

{
  "hotels": [
    {
      "hotelId": "H1001",
      "name": "Sunrise Grand Hotel",
      "location": {
        "address": "123 Beach Road",
        "city": "Miami",
        "country": "USA"
      },
      "contact": {
        "phone": "+1-555-123-4567",

```

```
"email": "info@sunrisegrand.com"
},
"rating": 4.5
},
{
  "hotelId": "H1002",
  "name": "Ocean View Resort",
  "location": {
    "address": "45 Ocean Drive",
    "city": "Los Angeles",
    "country": "USA"
  },
  "contact": {
    "phone": "+1-555-234-5678",
    "email": "contact@oceanview.com"
  },
  "rating": 4.2
},
{
  "hotelId": "H1003",
  "name": "Mountain Stay Inn",
  "location": {
    "address": "78 Hill Road",
    "city": "Denver",
    "country": "USA"
  },
  "contact": {
    "phone": "+1-555-345-6789",
    "email": "support@mountainstay.com"
  },
  "rating": 4.0
},
{
  "hotelId": "H1004",
  "name": "Royal Palace Hotel",
  "location": {
    "address": "12 King Street",
    "city": "London",
    "country": "UK"
  },
  "contact": {
    "phone": "+44-555-456-7890",
    "email": "info@royalpalace.co.uk"
  },
  "rating": 4.7
},
```

```

{
  "hotelId": "H1005",
  "name": "City Lights Hotel",
  "location": {
    "address": "88 Downtown Ave",
    "city": "New York",
    "country": "USA"
  },
  "contact": {
    "phone": "+1-555-567-8901",
    "email": "hello@citylights.com"
  },
  "rating": 4.3
}
}

```

Output:

```

PS C:\Users\2024503701\Desktop\FullStack\EX06> node server.js
Server is running at http://localhost:3000

```

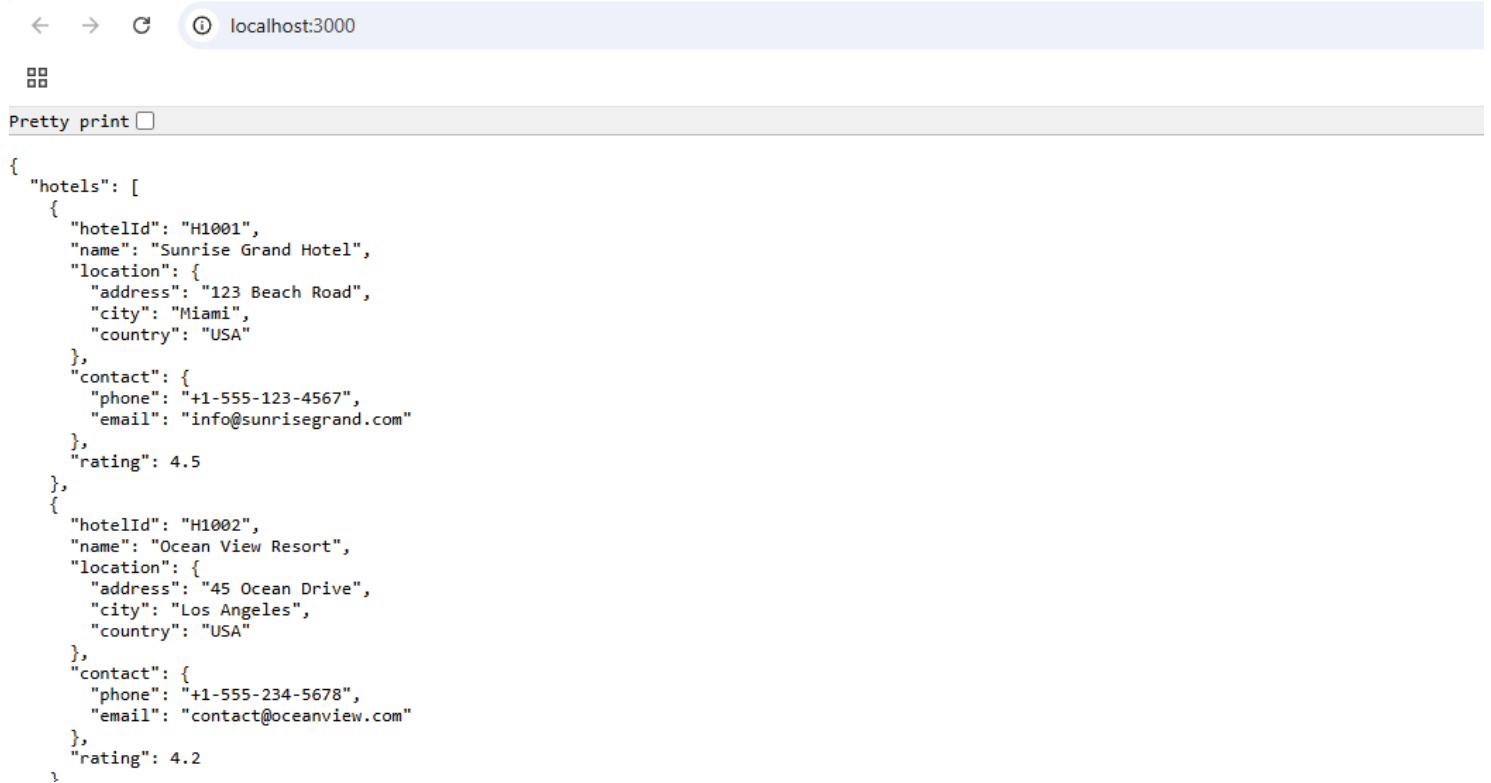
Figure(a) Starting the server

```

PS C:\Users\2024503701\Desktop\FullStack\EX06> node parse.js
{
  hotels: [
    {
      hotelId: 'H1001',
      name: 'Sunrise Grand Hotel',
      location: [Object],
      contact: [Object],
      rating: 4.5
    },
    {
      hotelId: 'H1002',
      name: 'Ocean View Resort',
      location: [Object],
      contact: [Object],
      rating: 4.2
    },
    {
      hotelId: 'H1003',
      name: 'Mountain Stay Inn',
      location: [Object],
      contact: [Object],
      rating: 4
    },
    {
      hotelId: 'H1004',
      name: 'Royal Palace Hotel',
      location: [Object],
      contact: [Object],
      rating: 4.7
    },
    {
      hotelId: 'H1005',
      name: 'City Lights Hotel',
      location: [Object],
      contact: [Object],
      rating: 4.3
    }
  ]
}

```

Figure(b) Parsing Console Output



```
{
  "hotels": [
    {
      "hotelId": "H1001",
      "name": "Sunrise Grand Hotel",
      "location": {
        "address": "123 Beach Road",
        "city": "Miami",
        "country": "USA"
      },
      "contact": {
        "phone": "+1-555-123-4567",
        "email": "info@sunrisegrand.com"
      },
      "rating": 4.5
    },
    {
      "hotelId": "H1002",
      "name": "Ocean View Resort",
      "location": {
        "address": "45 Ocean Drive",
        "city": "Los Angeles",
        "country": "USA"
      },
      "contact": {
        "phone": "+1-555-234-5678",
        "email": "contact@oceanview.com"
      },
      "rating": 4.2
    }
  ]
}
```

Figure(c)Running in localhost:3000

Exploration on npm commands Output:

```
PS C:\Users\2024503701\Desktop\FullStack\EX06> npm init
>>
This utility will walk you through creating a package.json file.
It only covers the most common items, and tries to guess sensible defaults.

See `npm help init` for definitive documentation on these fields
and exactly what they do.

Use `npm install <pkg>` afterwards to install a package and
save it as a dependency in the package.json file.

Press ^C at any time to quit.
package name: (ex06) json_parsing
version: (1.0.0)
description:
entry point: (parse.js)
test command:
git repository:
keywords:
author:
license: (ISC)
type: (commonjs)
About to write to C:\Users\2024503701\Desktop\FullStack\EX06\package.json:

{
  "name": "json_parsing",
  "version": "1.0.0",
  "description": "",
  "main": "parse.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

Is this OK? (yes)

npm notice
npm notice New minor version of npm available! 11.6.2 -> 11.7.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.7.0
npm notice To update run: npm install -g npm@11.7.0
npm notice
```

Figure(d)npm init

```
Wrote to C:\Users\2024503701\Desktop\FullStack\EX06\package.json:
>>
{
  "name": "json_parsing",
  "version": "1.0.0",
  "description": "",
  "license": "ISC",
  "author": "",
  "type": "commonjs",
  "main": "parse.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "keywords": []
}
```

Figure(e)npm init -y

```
Wrote to C:\Users\2024503701\Desktop\FullStack\EX06\package.json:
>>
{
  "name": "json_parsing",
  "version": "1.0.0",
  "description": "",
  "license": "ISC",
  "author": "",
  "type": "commonjs",
  "main": "parse.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "keywords": []
}

added 3 packages, and audited 4 packages in 1s

found 0 vulnerabilities
PS C:\Users\2024503701\Desktop\FullStack\EX06> █
```

Figure(f)npm install xml2js

```
PS C:\Users\2024503701\Desktop\FullStack\EX06> npm install express
>>

added 65 packages, and audited 69 packages in 4s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\2024503701\Desktop\FullStack\EX06> █
```

Figure(g)npm install express

```
added 65 packages, and audited 69 packages in 4s  
  
22 packages are looking for funding  
  run `npm fund` for details  
  
found 0 vulnerabilities  
found 0 vulnerabilities  
PS C:\Users\2024503701\Desktop\FullStack\EX06>
```

Figure(h)npm audit

Result:

Thus, the web server and JSON data handling system have been implemented successfully, and the output was generated by reading and parsing the JSON file, displaying the student information on the console and server.